

I Year II Semester

PHYSICS FOR ELECTRICAL SYSTEMS (EEE)**(AI Integrated)**

Course Code	Year/Sem	L	T	Pr	P	Credits
26APH02T	I Year II Sem	3	0	0	0	3

Prerequisites: Engineering Mathematics, Basic Electrical Engineering, Engineering Physics Fundamentals

Course Objectives

The objective of this course is :

1. To provide knowledge on wave optics concepts and their applications with basic AI-assisted optical analysis and sensing techniques.
2. To introduce the electrical properties of metals and semiconductors with simple AI-assisted conductivity and material analysis methods.
3. To equip students the fundamentals of quantum mechanics and quantum computing with basic AI-assisted secure communication concepts.
4. To familiarize students with dielectric and magnetic materials used in power systems along with basic AI-assisted monitoring and predictive analysis techniques.
5. To impart knowledge on electromagnetic field theory and its applications in power systems with simple AI-assisted EMI prediction and shielding concepts.

Course Content**UNIT I: Applied Wave Optics****9 hrs****Core Content:**

Interference: Principle of superposition and coherence, Interference, Types of interference, Interference in thin films (reflection geometry) and Newton's rings. Applications: Determination of wavelength and refractive index.

Diffraction: Introduction to diffraction; Fresnel and Fraunhofer diffraction. Fraunhofer diffraction due to single, double and N-slits (Diffraction Grating), Dispersive power and Resolving power of grating. Application: LiDAR for Autonomous Sensing in Electric Vehicles (EV).

Polarization: Types of polarization, Nicol's prism, Half-wave and Quarter-wave plates.

AI Integration:

AI-powered LiDAR integration for autonomous EV sensing and navigation. Python-based fringe pattern analysis for semiconductor wafer surface profiling.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn / basic DL models.

UNIT II: Electrical Properties of Metals and Semiconductors**8 hrs****Core Content:**

Classical free electron theory – merits and demerits, Quantum free electron theory – equation for electrical conductivity, Fermi-Dirac statistics, Fermi level and variation with temperature, Classification of solids – metals, insulators and semiconductors.

Semiconductors: Introduction, Fermi energy and Fermi level in intrinsic and extrinsic semiconductors, Expression for concentration of electrons in conduction band and holes in valence band, Electrical

conductivity, Expressions for drift and diffusion currents, Hall effect, Expression for Hall coefficient and its applications

AI Integration: AI-Assisted Prediction of Electrical Conductivity in Metals and Semiconductors, AI-Assisted Carrier Concentration and Conductivity Analysis in Semiconductors.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT III: Introduction to Quantum Mechanics & Quantum Computing

10 hrs

Core Content:

Introduction to quantum mechanics, de-Broglie concept (matter waves), Heisenberg's Uncertainty Principle, Wave function, Postulates of Quantum Mechanics, Schrödinger's time-independent wave equation, Particle in a one-dimensional potential well, Quantum tunnelling (basic idea).

Introduction to Quantum Computing: Classical bits vs Qubits. Concept of superposition and Entanglement, Basic Quantum Operations: Hadamard gate (idea of superposition), Pauli-X gate. Basic idea of quantum measurement. Applications (qualitative only): AI-Assisted Tunnel Diodes and cold emission of electrons, Basic idea of quantum cryptography and secure communication.

AI Integration: AI-Assisted Analysis of Tunnel Diode Characteristics using Quantum Tunnelling Concepts, AI-Assisted Quantum Key Distribution for Secure Communication

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT IV: Dielectrics and Magnetic Materials for Power Systems

9 hrs

Core Content:

Dielectric Materials: Introduction, Basic definitions, Types of polarizations, Internal fields in solids and the Clausius-Mossotti relation, Frequency dependence of the polarisation, Dielectric Loss. Applications (qualitative only): AI-Assisted Monitoring of Solid, Liquid and Gaseous Dielectrics in High Voltage Applications.

Magnetic Materials and Energy Conversion: Introduction and classification (Dia, Para, Ferro, Ferri, and Anti-ferro); Hysteresis and Eddy current losses in electrical machines; Skin effect; Soft & Hard magnetic materials. Applications (basic idea): Transformer cores, armatures, high-frequency inductors and permanent magnets for EV motors and wind generators.

AI Integration:

AI-based hysteresis loop analysis and eddy current loss estimation. AI-driven design of permanent magnets for EV motors.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT V: Electromagnetic Fields in Power Systems

9hrs

Core Content:

Introduction, wave equation in differential form in free space, Vector analysis, Divergence and curl of electric field and magnetic field (static), Gauss' divergence theorem and Stokes' theorem. Derivations of the four Maxwell equations and their significance. Maxwell's equations in vacuum, Energy transport and the Poynting Vector. Applications (qualitative only): Role of Maxwell's equations in Electromagnetic Interference (EMI), EMI in power converters (switching noise in IGBT/MOSFET-based inverters).

AI Integration: AI-Assisted EMI Prediction in IGBT/MOSFET-Based Power Converters, AI-Optimised Electromagnetic Shielding for Power Electronics.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

Course Outcomes (COs)

After completion of the course, the student will be able to

CO	Course Outcomes	Bloom's Level
CO1	Apply the principles of interference, diffraction, and polarization with basic AI-assisted optical analysis in engineering applications.	L3 (Apply)
CO2	Analyze the electrical properties of metals and semiconductors using basic AI-assisted material characterization and conductivity analysis techniques.	L4 (Analyze)
CO3	Apply the concepts of quantum mechanics and quantum computing with basic AI-assisted secure communication applications.	L3 (Apply)
CO4	Analyse dielectric and magnetic material behaviour in power systems using basic AI-assisted monitoring and predictive maintenance techniques.	L4 (Analyze)
CO5	Apply and analyze Maxwell's equations and electromagnetic field concepts for energy transport, AI-based EMI detection and mitigation in power electronic and smart grid systems.	L3 (Apply) L4 (Analyze)

CO – PO Articulation Matrix

Scale: 3 – Strong, 2 – Moderate, 1 – Slight, – No Correlation

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	3	–	–	–	–	–	–
CO2	3	3	2	2	3	–	–	–	–	–	–
CO3	3	3	2	2	2	–	–	–	–	–	–
CO4	3	3	3	2	3	–	–	–	–	–	–
CO5	3	3	2	3	3	–	–	–	–	–	–

Textbooks

1. Avadhanulu, Kshirsagar & Arun Murthy, Engineering Physics, S. Chand, 2019.
2. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford Press, 2015.

Reference Books

1. Shatendra Sharma & Jyotsna Sharma, Engineering Physics, Pearson, 2018.
2. S. O. Pillai, Solid State Physics, New Age International.
3. Nielsen & Chuang, Quantum Computation & Quantum Information, Cambridge.
4. Streetman & Banerjee, Solid State Electronic Devices, Pearson.
5. C. Kittel, Introduction to Solid State Physics, Wiley.
6. A. Von Hippel, Dielectrics and Waves, Wiley.
7. D. J. Griffiths, Introduction to Electrodynamics, Cambridge.
8. W. H. Hayt & J. A. Buck, Engineering Electromagnetics, McGraw Hill.

Web Resources & NPTEL Links

- NPTEL – nptel.ac.in
- MIT OpenCourseWare – ocw.mit.edu
- HyperPhysics – hyperphysics.phy-astr.gsu.edu
- Feynman Lectures – feynmanlectures.caltech.edu
- PhET Simulations – phet.colorado.edu
- Library of Congress – loc.gov/rr/scitech/selected-internet/physics.html

Recommended Free & Open-Source AI/Computational Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) – FREE on Google Colab for wave optics and EM field analysis
- IBM Qiskit (free, open-source) – For quantum computing and quantum cryptography simulations on Google Colab
- scikit-learn (free, open-source) – For semiconductor property prediction and EMI data analysis
- OpenEMS (free, open-source) – For 3D electromagnetic field simulation and power system analysis
- PhET Interactive Simulations (free) – For wave optics, quantum mechanics and semiconductors

I Year II Semester

Course Code	Title	L	T	P	C
26APH02P	PHYSICS FOR ELECTRICAL SYSTEMS LAB (AI INTEGRATED)	0	0	3	1.5

Prerequisites: Basic Physics and Mathematics (Class XII level); no concurrent prerequisite

Course Objectives

1. To determine the optical properties of materials — dispersive power, thickness, radius of curvature, and wavelength of light — using prism, wedge, Newton's-rings, and diffraction-grating experiments.
2. To investigate laser and quantum phenomena by measuring laser wavelength using diffraction and by estimating Planck's constant through the photoelectric effect.
3. To study the electrical, magnetic, dielectric, and semiconductor properties of materials — including B–H curves, Hall effect, energy gap, resistivity, and the Curie temperature/dielectric behaviour of ferroelectric materials — that underpin electrical and electronic devices and machines.
4. To analyze magnetic field distributions—verifying the magnetic inverse square law and studying axial field variations via the Stewart & Gee method—for applications in electrical and power systems.
5. To determine fundamental electromagnetic properties, including the electron charge-to-mass ratio (e/m) and the self-inductance of circuit components, using standard electrical measurement instruments.

List of Experiments (Any ten experiments need to be performed)

Part A – Regular Experiments

1. Determination of the dispersive power of the prism.
2. Determination of the thickness of a thin object by the wedge method
3. Determination of the radius of curvature of the lens by Newton's rings
4. Determination of wavelengths of various colours of mercury spectrum using a diffraction grating in the normal incidence method
5. Determination of wavelength of a LASER using a diffraction grating
6. Determination of the resistivity of a semiconductor by the four-probe method.
7. Determination of the energy gap of a semiconductor using a p-n junction diode.
8. Determination of Hall voltage and Hall coefficient of a given semiconductor using the Hall effect.
9. Estimation of Planck's constant using the photoelectric effect
10. Determination of e/m of an electron by J.J. Thomson method.
11. Determination of the Curie temperature and dielectric constant of a ferroelectric material
12. Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
13. Determination of magnetic field along the axis of a current-carrying circular coil by Stewart & Gee's Method
14. Verification of the inverse square law for a magnetic field using a Hall probe.
15. Measurement of self-inductance of a coil using LCR meter and resonance method.

Part B — AI Integration Lab Exercises

Any two (2) of the following AI exercises must be completed using Python / Google Colab or an equivalent AI/LLM tool.

1. **Optics Data Automation (Python / Matplotlib):** input raw angle-of-deviation data from the prism experiment; fit a polynomial to the deviation-vs-angle curve; extract the angle of minimum deviation and compute the dispersive power; overlay the fitted curve on the experimental data points.
2. **Fringe Analysis using OpenCV (Python / Google Colab):** import Newton's-ring or wedge-fringe images; apply Gaussian blur and Canny edge detection to locate fringe centres; compute fringe diameters automatically; calculate radius of curvature or thickness and compare with manual measurements.
3. **Planck's Constant Estimation (Python / NumPy):** input stopping-potential vs. frequency data from the photoelectric experiment; apply linear regression (V_0 vs. ν); extract the slope to compute h and the y-intercept to find the work function; plot the regression line and quantify percentage error against the accepted value 6.626×10^{-34} J·s.

AI Integration:

AI integration adds two to three exercises after the regular experiments: (i) Python automation of prism deviation curve-fitting and dispersive-power extraction using Matplotlib/NumPy; (ii) OpenCV-based fringe-image analysis for Newton's rings and the wedge method, automating diameter measurement and radius-of-curvature computation; (iii) linear regression on photoelectric-effect data to extract Planck's constant and the work function — reinforcing the connection between the physical experiment and quantum theory, and giving EEE students hands-on practice applying programming, EM-field simulation, and data-analysis skills to experimental physics and electrical engineering problems.

Free / Open-Source AI Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) on Google Colab (free) — wave-optics automation, fringe analysis, photoelectric regression, and EM field analysis
- scikit-learn (free, open-source) — semiconductor property prediction and EMI data analysis
- OpenEMS (free, open-source) — 3D electromagnetic field simulation and power-system analysis
- PhET Interactive Simulations (free) — wave optics, quantum mechanics, and semiconductors: phet.colorado.edu
- Virtual Labs — IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences — online simulation of optics, semiconductor, and EM experiments
- HyperPhysics (free) — concept reference for optics, quantum, and EM theory: hyperphysics.phy-astr.gsu.edu

Web Resources

- HyperPhysics — hyperphysics.phy-astr.gsu.edu
 - PhET Simulations — phet.colorado.edu
 - NPTEL — npTEL.ac.in
 - Virtual Labs (IIT Bombay) — iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences
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Course Outcomes (COs)

After successful completion of this lab course, the student will be able to:

CO	Course Outcomes	Bloom's Level
CO1	Compute key optical properties of materials—including dispersive power, thin-film thickness, lens curvature radius, and light wavelengths—by operating prism, interference, and diffraction setups.	L3 (Apply)
CO2	Analyze quantum photo-emission dynamics and laser diffraction phenomena to estimate Planck's constant and determine laser wavelengths	L3–L4 (Apply, Analyze)
CO3	Evaluate the electrical, magnetic, dielectric, and semiconductor characteristics of materials—such as B–H hysteresis, Hall coefficient, energy band gap, electrical resistivity, and ferroelectric Curie transition—for device applications.	L4 (Analyze)
CO4	Evaluate magnetic field propagation dynamics by verifying the magnetic inverse square law and mapping axial magnetic field variations using the Stewart & Gee setup.	L4–L5 (Analyze, Evaluate)
CO5	Determine fundamental electromagnetic parameters, including the specific charge (e/m) of an electron and the self-inductance of electrical components, using standard measuring instruments.	L3–L4 (Evaluate, Create)

CO – PO / PSO Articulation Matrix (NBA)

Scale: 3 – Strong | 2 – Moderate | 1 – Slight | – No Correlation

CO	Bloom's Lvl	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	L3	3	2	2	3	2	–	–	–	–	–	–
CO2	L3–L4	3	3	2	3	3	–	–	–	–	–	–
CO3	L4	3	3	2	3	2	–	–	–	–	–	–
CO4	L3–L4	3	2	2	3	2	–	–	–	–	–	–
CO5	L5–L6	2	3	3	3	3	–	–	–	–	–	–

Recommended Free & Open-Source AI / Computational Tools

Python (NumPy, SciPy, OpenCV, Matplotlib) on Google Colab (free) – optics automation, fringe analysis, photoelectric regression, EM simulations

scikit-learn (free, open-source) – semiconductor data analysis and signal classification

OpenEMS (free, open-source) – electromagnetic field simulation and antenna design

PhET Interactive Simulations (free) – wave optics, semiconductors, and quantum phenomena: phet.colorado.edu

Virtual Labs – IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences – online simulation of optics, semiconductor and EM experiments

HyperPhysics (free) – concept reference for optics, quantum and EM theory: hyperphysics.phy-astr.gsu.edu

LibreOffice Calc (free, open-source) – standardised data recording templates for all experiments
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Note: Lab record to be submitted for every experiment at the next session. AI exercises to be submitted as Google Colab (.ipynb) notebooks.

Reference Books

1. M. N. Avadhanulu & P. G. Kshirsagar, A Textbook of Engineering Physics, S. Chand & Company, 2019.
2. H. K. Malik & A. K. Singh, Engineering Physics, Tata McGraw-Hill, 2010.
3. R. K. Gaur & S. L. Gupta, Engineering Physics, Dhanpat Rai Publications, 8th Edition, 2012.
4. B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2012.
5. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford University Press, 2015.